Input: Sample access to two pure quantum states $|\varphi\rangle$ and $|\psi\rangle$; the desired additive error $\varepsilon \in (0,1)$.

Output: ε-estimates of $\mathrm{T}(|\varphi\rangle,|\psi\rangle)$ and $\mathrm{F}(|\varphi\rangle,|\psi\rangle)$ with probability at least $2/3$.

1: $t \coloneqq \Theta(\log(1/\varepsilon))$.

2: Let $\mathcal{A}^{U_1,U_2} \coloneqq \mathsf{QPE}^{U_1U_2}_{\frac{1}{10},\frac{\varepsilon}{\pi}}$ denote the quantum circuit for phase estimation of U_1U_2.

3: Prepare the quantum state $\rho \coloneqq \mathsf{Samplize}^{\mathsf{pure}}_{\frac{1}{10}}\langle \mathcal{A}^{U_1,U_2}\rangle\big[|\varphi\rangle,|\psi\rangle\big]\big(|0\rangle\langle 0|^{\otimes t}\otimes|\varphi\rangle\langle\varphi|\big)$.

4: Let $b_1,b_2,\ldots,b_t \in \{0,1\}$ be the outcome of measuring the first t qubits of ρ in the computational basis, and let $\tilde{\gamma} \coloneqq \sum_{j=1}^{t} b_j 2^{-j} \in [0,1)$ be the estimate of the phase.

5: **return** $|\cos(\pi\tilde{\gamma}-\frac{\pi}{2})|$ and $|\sin(\pi\tilde{\gamma}-\frac{\pi}{2})|$ as the estimates of $\mathrm{T}(|\varphi\rangle,|\psi\rangle)$ and $\mathrm{F}(|\varphi\rangle,|\psi\rangle)$, respectively.